

Oil Well #807

Production Performance Analysis

Python time-series analysis | Production decline | Water burden | Pressure depletion

ANALYSIS SCOPE

2,939 daily observations | 2013-01-01 to 2021-01-18 | Nine source fields

Observed performance, data-quality checks, historical decline screening, and review priorities

No unsupported reserves estimate, economic-limit conclusion, or EOR prescription

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2,939

Daily observations

49 to 5

Oil rate (m3/day)

89.8%

Oil-rate decline

51,798 m3

Observed oil volume

29% to 70%

Water cut

214 to 100

Pressure (atm)

93.10%

Hours utilization

1

Zero-oil observation

What the data supports

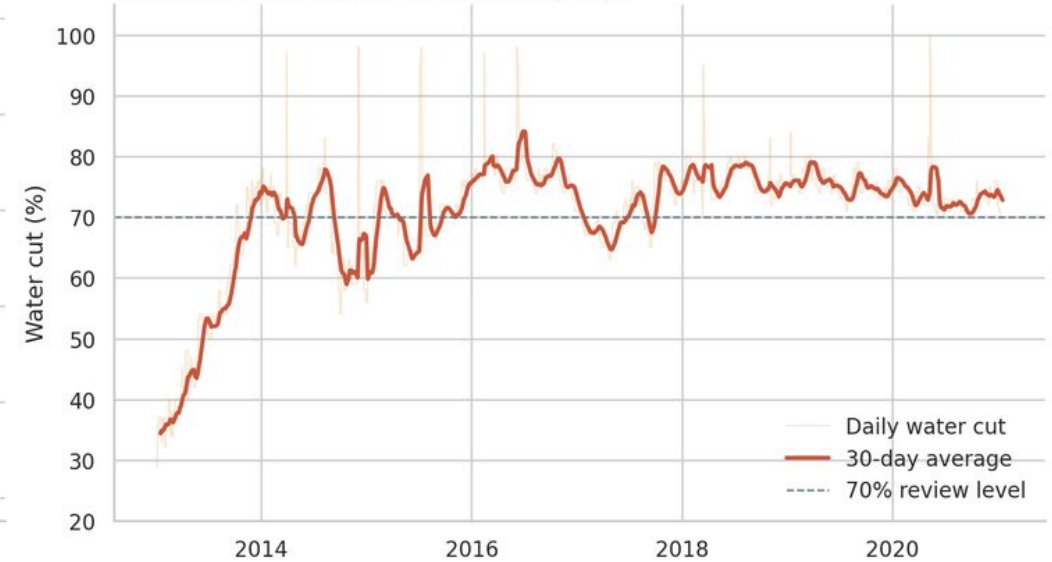
- Material oil-rate decline occurred alongside reservoir-pressure depletion and sustained produced-water burden.
- The operating-hours series indicates that downtime and lift constraints should be separated from natural decline.
- The history supports engineering review priorities, but it does not identify a shutdown cause or prescribe an EOR method.

Well #807 | Production Performance

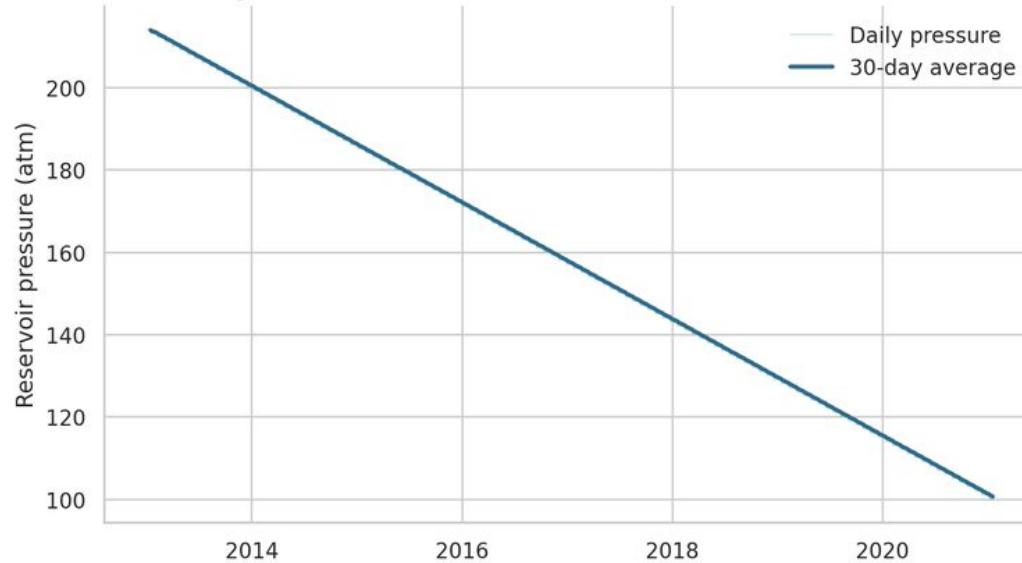
Oil rate declined materially across the history



Water burden became persistently high



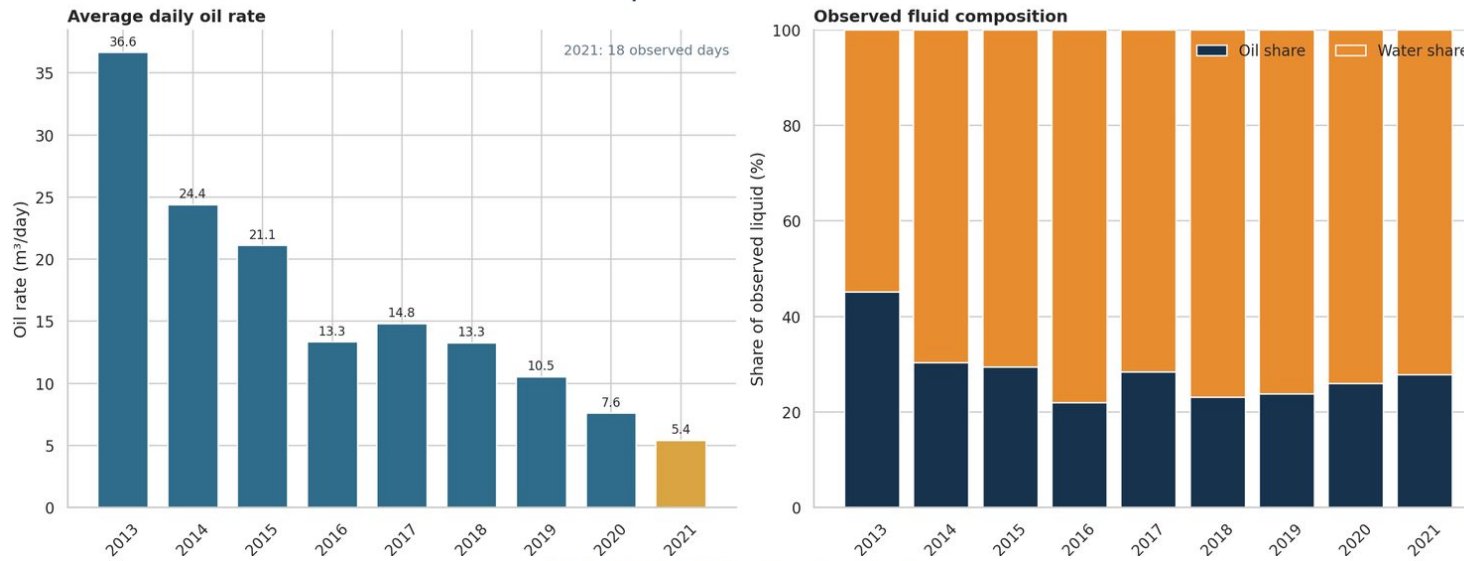
Reservoir pressure decreased from 214 to 100 atm



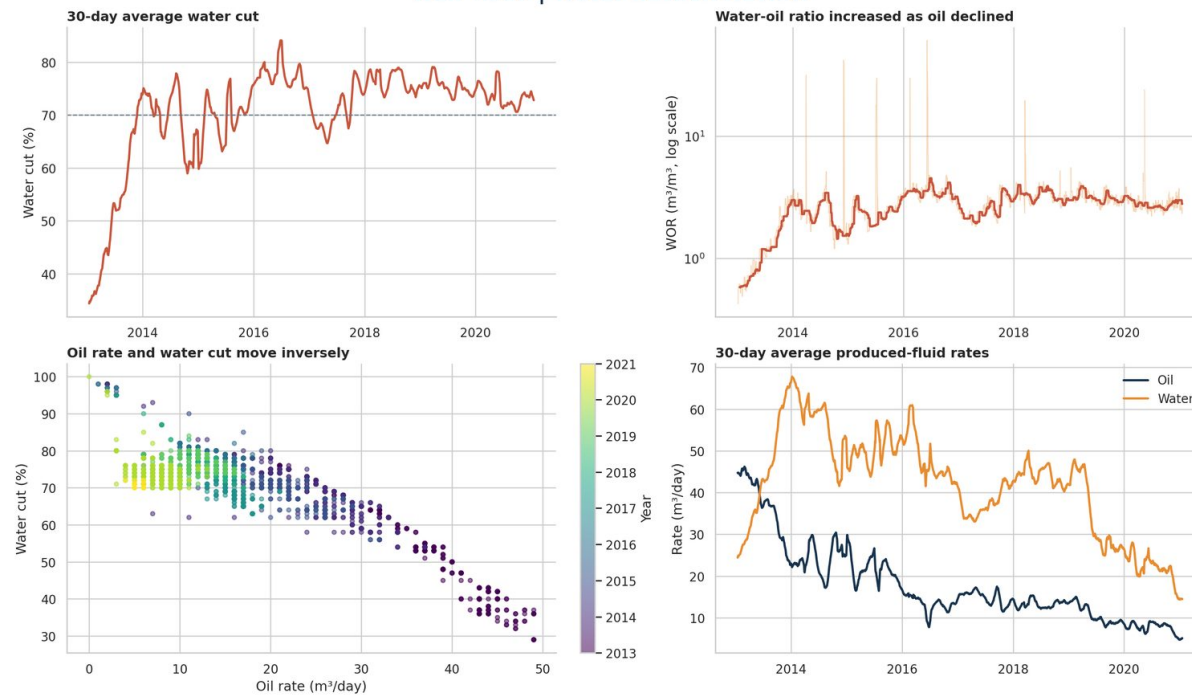
Operating availability weakened late in the history



Well #807 | Annual Production Profile

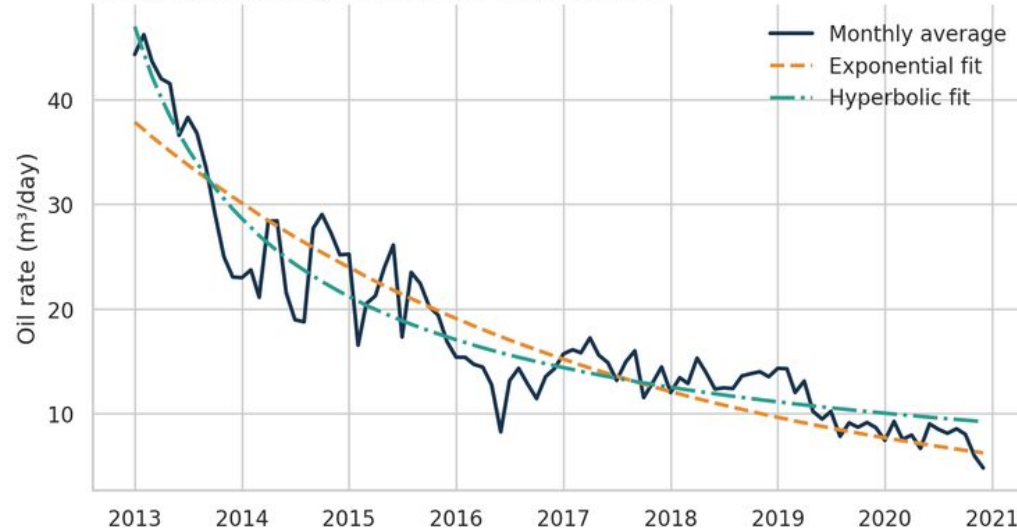


Well #807 | Water Burden Review

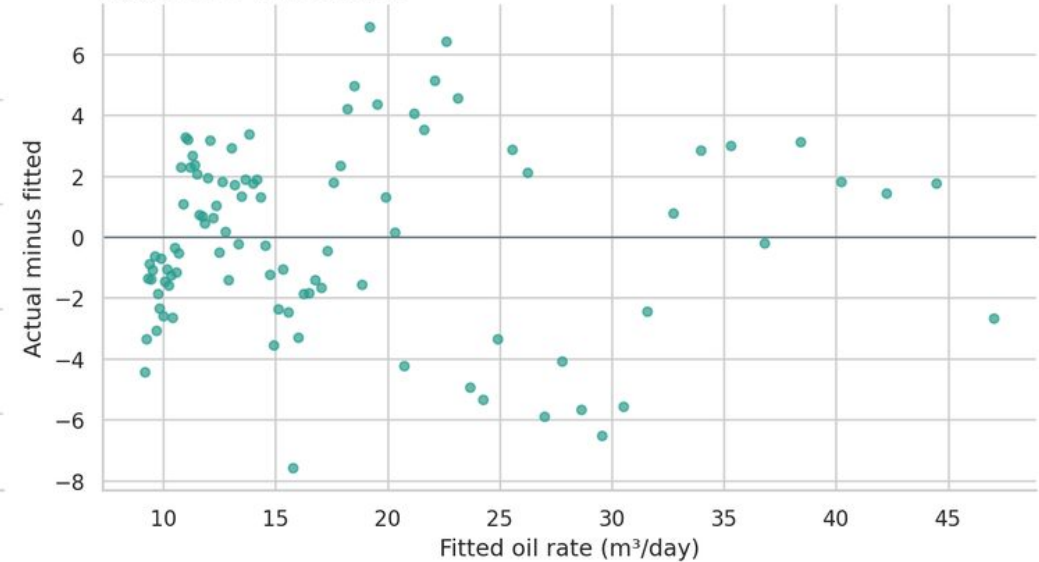


Well #807 | Decline-Curve Screening

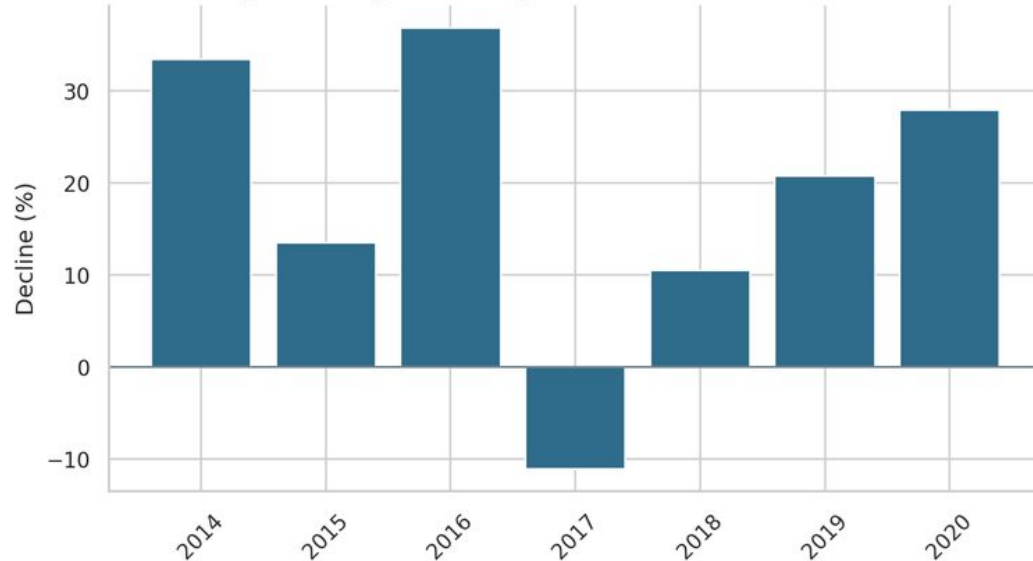
Historical monthly oil rate and fitted curves



Hyperbolic-fit residuals



Year-over-year change in average oil rate

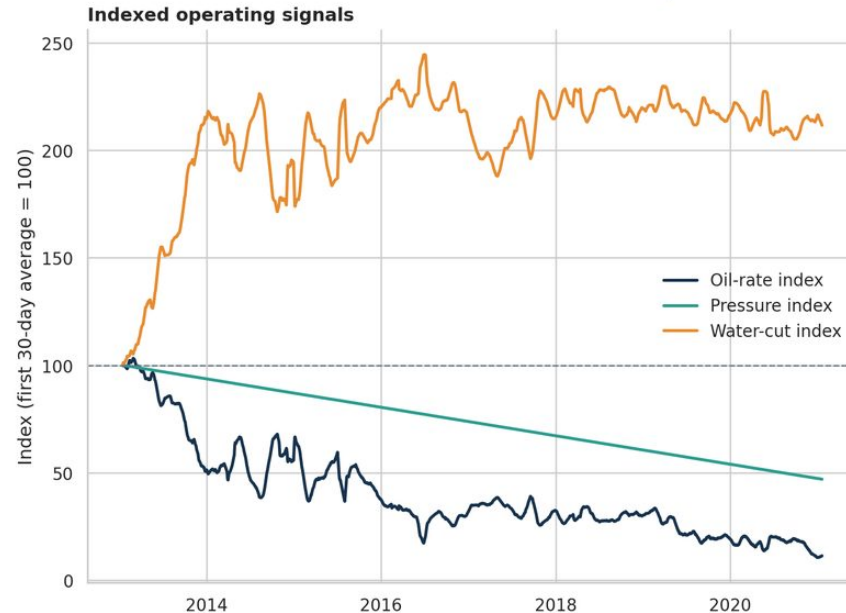


SCREENING RESULT

Exponential R^2 : 0.849
Hyperbolic R^2 : 0.902

The hyperbolic curve describes the observed history better, but neither fit is a reserves estimate. Economic limits require price, cost, downtime, intervention, and engineering assumptions.

Well #807 | Intervention Review Priorities



PRIORITY REVIEW

- 1 Diagnose water source and water-handling cost
- 2 Separate natural decline from downtime and lift constraints
- 3 Review artificial-lift settings and intervention history
- 4 Add economics before defining an economic limit
- 5 Screen EOR only with field and reservoir data

Required before economic-limit, reserves, or EOR decisions

- Failure, shutdown, workover, and intervention history.
- Artificial-lift settings, well tests, completion intervals, and water-source diagnostics.
- Oil price, lifting cost, water-disposal cost, intervention cost, and abandonment liability.
- Offset-well performance, injection history, PVT, rock properties, and reservoir simulation.